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Unit 2 : Session 1 : SLO 2 | Inserting an Element into an Array

I. Pseudocode " Append VAL

- Step 1: Set $\text{upper_bound} = N + 1$ (the array grows by one; new upper bound)
- Step 2: Set $A[N + 1] = \text{VAL}$ (place the value at the new last position)
- Step 3: EXIT

II. Pseudocode " Insert VAL at position POS

- Step 1: [INITIALIZATION] SET $I = N$
- Step 2: Repeat Steps 3 and 4 while $I \geq \text{POS}$
- Step 3: SET $A[I + 1] = A[I]$ (shift element right)
- Step 4: SET $I = I - 1$
- [END OF LOOP]
- Step 5: SET $N = N + 1$
- Step 6: SET $A[\text{POS}] = \text{VAL}$
- Step 7: EXIT
- Note: shifting starts from the last element downward so no element is overwritten before being copied.